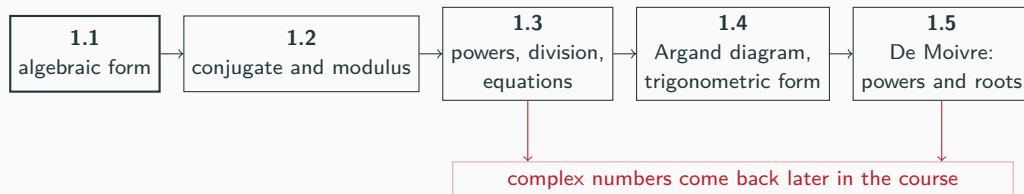


Linear Algebra

1 Complex numbers

José Firmino Aguilar Madeira

Chapter map



Exercises: Sheet 1, Ex. 1–14 (original numbering).

1.1 Algebraic form

Why $\mathbb{C}$?

In $\mathbb{R}$ some equations have no solution

$$x^2 + 1 = 0 \quad (\text{no real number has } x^2 = -1).$$

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In geometry too

The 90° rotation of the plane:

$$(x, y) \mapsto (-y, x).$$

Is any direction preserved, $(-y, x) = \lambda(x, y)$?

$$-y = \lambda x, \quad x = \lambda y \Rightarrow (1 + \lambda^2)y = 0.$$

With λ real, only $(x, y) = (0, 0)$: **no direction is preserved.**

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The solution

Add to $\mathbb{R}$ a number i with

$$i^2 = -1.$$

Then $x^2 + 1 = 0$ has the solutions i and $-i$, and the condition $1 + \lambda^2 = 0$ from the rotation also has a solution: $\lambda = \pm i$.

We return to this rotation at the end of the chapter.

Definition

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A **complex number** is an expression $z = a + bi$, with $a, b \in \mathbb{R}$ and $i^2 = -1$.

$$\mathbb{C} = \{ a + bi : a, b \in \mathbb{R} \}.$$

$a = \operatorname{Re} z$ is the **real part**; $b = \operatorname{Im} z$ is the **imaginary part** (it is a real number).

- $b = 0$: $z = a$ is real ($\mathbb{R} \subset \mathbb{C}$). $a = 0, b \neq 0$: $z = bi$ is **purely imaginary**.
- **Equality:** $a + bi = c + di \iff a = c$ and $b = d$.

Caution

Do not write $i = \sqrt{-1}$: it leads to errors such as $\sqrt{-1} \sqrt{-1} = \sqrt{1} = 1$, whereas $i \cdot i = -1$.

Formally, $\mathbb{C}$ is $\mathbb{R}^2$ with the product $(a, b)(c, d) = (ac - bd, ad + bc)$; i is the pair $(0, 1)$. With addition and this product, $\mathbb{C}$ is a field.

Sum and product

Compute as with polynomials in i , using $i^2 = -1$:

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i.$$

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Example. $z_1 = 3 + 2i$, $z_2 = 1 - i$:

$$z_1 + z_2 = 4 + i, \quad z_1 z_2 = 3 - 3i + 2i - 2i^2 = 5 - i.$$

The usual properties still hold: commutativity, associativity, distributivity.

- **Ex. 1** operations in algebraic form.
- **Ex. 2** a system in the unknowns $\operatorname{Re} z$ and $\operatorname{Im} z$.
- **Ex. 3** the conditions ‘real’ ($\operatorname{Im} = 0$) and ‘purely imaginary’ ($\operatorname{Re} = 0, \operatorname{Im} \neq 0$).

1.2 Conjugate and modulus

Definitions

Let $z = a + bi$.

- **Conjugate:** $\bar{z} = a - bi$ (changes the sign of the imaginary part).
- **Modulus:** $|z| = \sqrt{a^2 + b^2}$ (a real number ≥ 0).

Conjugate and modulus

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$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2 \text{ is always real.}$$

Example. $z = 3 + 4i$: $\bar{z} = 3 - 4i$, $|z| = \sqrt{9 + 16} = 5$, $z\bar{z} = 25$.

For any $z, z_1, z_2 \in \mathbb{C}$:

Conjugate

- $\overline{\overline{z}} = z$
- $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
- $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$
- $z + \overline{z} = 2 \operatorname{Re} z, \quad z - \overline{z} = 2i \operatorname{Im} z$
- z is real $\iff z = \overline{z}$

Modulus

- $|z| = 0 \iff z = 0$
- $|\overline{z}| = |z|$
- $|z_1 z_2| = |z_1| |z_2|$
- $|z_1 + z_2| \leq |z_1| + |z_2|$

In general $|z_1 + z_2| \neq |z_1| + |z_2|$: for example, $|1 + (-1)| = 0$.

- **Ex. 4** moduli.
- **Ex. 5** check properties of the conjugate on an example.
- **Ex. 6** use $z\bar{z} = |z|^2$ to factorise $a^2 + b^2$ in $\mathbb{C}$.

1.3 Powers of i , division and equations

Powers of i

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \quad \dots$$

The powers repeat every 4. If $n = 4k + r$, with $r \in \{0, 1, 2, 3\}$ the remainder of dividing n by 4:

$$i^n = i^r.$$

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$$i^n = i^r.$$

Example. $2026 = 4 \cdot 506 + 2$, so $i^{2026} = i^2 = -1$.

To divide, multiply numerator and denominator by the **conjugate of the denominator**:

$$\frac{z_1}{z_2} = \frac{z_1 \overline{z_2}}{z_2 \overline{z_2}} = \frac{z_1 \overline{z_2}}{|z_2|^2} \quad (z_2 \neq 0).$$

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Example. $\frac{5-i}{1-i} = \frac{(5-i)(1+i)}{(1-i)(1+i)} = \frac{5+5i-i-i^2}{2} = \frac{6+4i}{2} = 3+2i.$

This confirms the product example: $(3+2i)(1-i) = 5-i.$

Inverse: $\frac{1}{z} = \frac{\overline{z}}{|z|^2}$; for example $\frac{1}{3+4i} = \frac{3-4i}{25}.$

Quadratic equations

In $\mathbb{C}$, the quadratic formula always works. If $\Delta < 0$, $\sqrt{\Delta} = i\sqrt{-\Delta}$.

Example. $z^2 - 2z + 5 = 0$: $\Delta = 4 - 20 = -16$, $\sqrt{\Delta} = 4i$,

$$z = \frac{2 \pm 4i}{2} = 1 \pm 2i.$$

Check: $(1 + 2i)^2 - 2(1 + 2i) + 5 = 1 + 4i - 4 - 2 - 4i + 5 = 0$.

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The two roots are **conjugates**. This is no coincidence: the coefficients are real.

Fundamental Theorem of Algebra

A polynomial of degree $n \geq 1$ with coefficients in $\mathbb{C}$ has exactly n roots in $\mathbb{C}$, each counted as many times as its multiplicity.

Real coefficients

If the coefficients are *real* and z is a root, then $\bar{z}$ is also a root: the non-real roots come in conjugate pairs.

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Example. $x^3 - 1 = (x - 1)(x^2 + x + 1)$: roots 1 and $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ (3 roots; one conjugate pair).

This fact becomes important again later in the course.

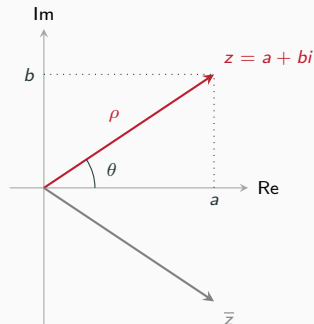
- **Ex. 7** powers of i .
- **Ex. 8** divisions in the form $a + bi$.
- **Ex. 9** equations in $\mathbb{C}$. In b), start by factorising.

1.4 Argand diagram and trigonometric form

Argand diagram

The complex number $z = a + bi$ is represented by the point (a, b) of the plane: its **image point**.

- horizontal axis: *real axis*;
- vertical axis: *imaginary axis*;
- $|z|$ is the distance from the image point to the origin;
- $\bar{z}$ is the reflection in the real axis;
- $|z_1 - z_2|$ is the **distance** between the image points of z_1 and z_2 .



Trigonometric form

With $\rho = |z|$ and θ the angle between the image point and the positive real axis:

$$a = \rho \cos \theta, \quad b = \rho \sin \theta, \quad z = \rho(\cos \theta + i \sin \theta) = \rho \operatorname{cis} \theta.$$

θ is an **argument** of z (defined up to multiples of 2π ; for $z = 0$ it is not defined). **Principal argument** (course convention): $\theta \in [0, 2\pi[$.

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Example. $z = 1 + i\sqrt{3}$: $\rho = \sqrt{1+3} = 2$, $\cos \theta = \frac{1}{2}$, $\sin \theta = \frac{\sqrt{3}}{2} \Rightarrow z = 2 \operatorname{cis} \frac{\pi}{3}$.

One also writes $\rho e^{i\theta}$ (Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$).

Mind the quadrant

$\tan \theta = \frac{b}{a}$ **is not enough**: $1 + i$ and $-1 - i$ have the same quotient $b/a = 1$.

Example. $z = -1 - i$: $\rho = \sqrt{2}$, $\cos \theta = -\frac{\sqrt{2}}{2}$, $\sin \theta = -\frac{\sqrt{2}}{2}$.

The image point is in the 3rd quadrant, so $\theta = \frac{5\pi}{4}$ (not $\frac{\pi}{4}$):

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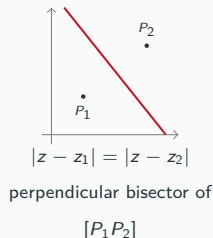
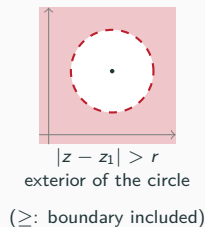
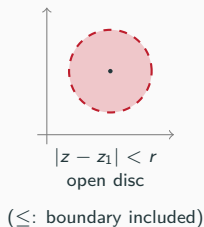
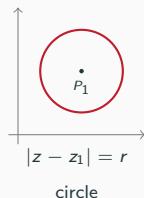
$$-1 - i = \sqrt{2} \operatorname{cis} \frac{5\pi}{4}.$$

Rule of thumb

Compute ρ ; use $\cos \theta = a/\rho$ and $\sin \theta = b/\rho$ (both signs), or sketch the image point.

Regions in the plane

With $z = x + yi$, z_1 fixed with image point P_1 , $r > 0$:



Why: $|z - z_1|$ is the distance from z to P_1 . With $z_1 = a + bi$:

$$|z - z_1| = r \iff (x - a)^2 + (y - b)^2 = r^2.$$

Also: $\operatorname{Re} z = a$ (vertical line), $\operatorname{Im} z = b$ (horizontal line), $\arg(z - z_1) = \theta$ (ray starting at P_1 , without P_1).

- **Ex. 10** trigonometric form (mind the quadrant in f , g , h)).
- **Ex. 11** algebraic form.
- **Ex. 12** regions: find the centre and radius; boundary included or not.
- **Ex. 13** rewrite in terms of x and y and recognise the curve.

1.5 De Moivre's formulae

Product and quotient in trigonometric form

Let $z_1 = \rho_1 \operatorname{cis} \theta_1$ and $z_2 = \rho_2 \operatorname{cis} \theta_2$.

$$z_1 z_2 = \rho_1 \rho_2 \operatorname{cis}(\theta_1 + \theta_2), \quad \frac{z_1}{z_2} = \frac{\rho_1}{\rho_2} \operatorname{cis}(\theta_1 - \theta_2) \quad (z_2 \neq 0).$$

To multiply: multiply the moduli and add the arguments. (This comes from $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$.)

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Example. $2 \operatorname{cis} \frac{\pi}{6} \cdot 3 \operatorname{cis} \frac{\pi}{3} = 6 \operatorname{cis} \frac{\pi}{2} = 6i$.

In algebraic form: $(\sqrt{3} + i)\left(\frac{3}{2} + \frac{3\sqrt{3}}{2}i\right) = 6i$. The trigonometric form is quicker.

De Moivre's formula (powers)

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For $z = \rho \operatorname{cis} \theta$ and $n \in \mathbb{Z}$ ($z \neq 0$ if $n < 0$):

$$z^n = \rho^n \operatorname{cis}(n\theta).$$

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Example. $(1 + i)^8$: $1 + i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$, so

$$(1 + i)^8 = (\sqrt{2})^8 \operatorname{cis} \frac{8\pi}{4} = 16 \operatorname{cis}(2\pi) = \mathbf{16}.$$

In algebraic form this would take 7 products.

n-th roots

If $w = \rho \operatorname{cis} \theta \neq 0$, the equation $z^n = w$ has exactly n solutions:

$$z_k = \sqrt[n]{\rho} \operatorname{cis} \frac{\theta + 2k\pi}{n}, \quad k = 0, 1, \dots, n-1.$$

Their image points are the vertices of a regular n -sided polygon centred at the origin.

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Example. $z^3 = 1 = \operatorname{cis} 0$: $z_k = \operatorname{cis} \frac{2k\pi}{3}$, that is,

$$1, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

These are the roots of $x^3 - 1$ that we found in 1.3, now without factorising.

- **Ex. 14** solve $z^4 - 8z = 0$. Hint: factor out z and use the cube roots.

Multiplying $z = x + yi$ by $\operatorname{cis} \theta$ rotates the image point through an angle θ (the modulus does not change):

$$\operatorname{cis} \theta (x + yi) = (x \cos \theta - y \sin \theta) + (x \sin \theta + y \cos \theta) i.$$

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In plane coordinates, this is the **rotation through angle θ** :

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LINEAR ALGEBRA IN PRACTICE: multiplying is rotating

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For $\theta = \frac{\pi}{2}$ (multiplying by i): $i(x + yi) = -y + xi$, that is, $(x, y) \mapsto (-y, x)$ — the rotation from the start. **Rotating by 90° and multiplying by i are the same operation**: that is why the question at the start only has an answer with $\lambda = \pm i$.

Used in computer graphics (rotating shapes) and in electrical engineering (phasors in AC circuits).

Concept	How to obtain it / what it says
Algebraic form	$z = a + bi$, $a = \operatorname{Re} z$, $b = \operatorname{Im} z$, $i^2 = -1$
Product	$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$
Conjugate, modulus	$\bar{z} = a - bi$, $ z = \sqrt{a^2 + b^2}$, $z\bar{z} = z ^2$
Division	multiply top and bottom by $\bar{z}_2$
Powers of i	$i^n = i^r$, $r = \text{remainder of } n \text{ divided by } 4$
Polynomials	degree n : n roots in $\mathbb{C}$; real coefficients: conjugate pairs
Trigonometric form	$z = \rho \operatorname{cis} \theta$, $\rho = z $, $\theta \in [0, 2\pi[$ (check the quadrant)
Distance	$ z_1 - z_2 = \text{distance between the image points}$
Regions	$ z - z_1 = r$ circle; $< r$ / $> r$ interior / exterior
Product	multiply the moduli and add the arguments (quotient: divide, subtract)
De Moivre	$(\rho \operatorname{cis} \theta)^n = \rho^n \operatorname{cis}(n\theta)$
Roots	$z^n = \rho \operatorname{cis} \theta$: $\sqrt[n]{\rho} \operatorname{cis} \frac{\theta + 2k\pi}{n}$, $k = 0, \dots, n-1$

KEY TAKEAWAYS — Chapter 1

1. The complex numbers add to the reals a number whose square is -1 ; with them, every polynomial has roots.
2. Multiplying a complex number by its conjugate gives a real number, and that is what makes division possible.
3. A complex number is a point in the plane: the algebraic form gives its coordinates, the trigonometric form gives its distance to the origin and its angle.
4. Multiplying complex numbers means multiplying lengths and adding angles; that is why powers and roots are simple in trigonometric form.
5. Rotating in the plane and multiplying by a complex number are the same operation; that is why complex numbers reappear throughout the course.

Next (Ch. 2): vector spaces — $\mathbb{R}^n$, $\mathbb{C}^n$ and polynomials as examples of the same concept.

References (full list: course bibliography)

- Churchill, R. V., Brown, J. W., *Complex Variables and Applications*, McGraw-Hill — chapter 1 (complex numbers).
- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese).
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill.

Resources — Sheet 1 (exercises 1–14) and its solutions.